

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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Abstract

How could activity distributed across cortex support one experienced situation? We separate three requirements: *coherence*, meaning collective phase order; *compatibility*, meaning that local descriptions agree where their domains overlap; and *self-representation*, meaning that the resulting state carries an accurate reconstruction of itself. The first supplies neither of the others, and four results establish that without any cortical hypothesis. Synchrony constrains content only through encoders shared on overlaps, and then by the resultant of the patch the bound is read on rather than of the whole population. The resultant does not determine the state: a winding field and a uniform one are stationary at one coupling with resultants spanning nearly the range of that observable. A contraction with a unique fixed point can discard its input, so accuracy across a family of states requires distinguishable codes. And where the encoding is a region's restriction, that resolution is demanded of the region before any readout is chosen. Under stated hypotheses, compatible local states then glue to a unique global state that an accurate contracting reconstruction fixes uniquely, and installed energy bounds the mean coupling a declared actuator supplies, so a coherent stationary branch requires it

to exceed $2D/\kappa$, with D the phase diffusion and κ set by the mode prices; energy is not the scarce side of that condition for a macroscopic substrate, which leaves it a statement about κ , derived here for neither cortex nor a processor. Lean 4 checks these results and their composition, with the eight connecting hypotheses as explicit arguments and no axioms beyond its libraries'. Further controls bound what observation can decide: distinct mechanisms share stationary phase distributions, and a coupling rule descending a squared-drift objective holds coherence while losing the structure it was meant to acquire. One prediction needs no absolute calibration: onset delay scales as the square root of the rate at which coupling crosses threshold, measured here as $\Delta K \propto v^{0.444}$, which a within-subject emergence-from-anaesthesia design can test. We propose the glued state and its internal reconstruction as correlates of unity and a minimal self, and an extracellular-field hypothesis linking awakening to rising coupling. Testing it requires calibrated geometry, resolved phase dynamics, validated decoders and a spatial observable beside the resultant.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful coordination would still leave a representational question: what makes the coordinated activity a description of one situation, with a perspective on that situation?

We distinguish three requirements. *Coherence* concerns collective phase order. *Compatibility* concerns whether local descriptions agree where their

domains overlap. *Self-representation* concerns whether the resulting state carries an accurate internal reconstruction of itself. We propose the latter two relations as formal correlates of conscious unity and a minimal self; their interpretation requires evidence beyond the phase model.

Unity has been approached differently elsewhere: integrated information theory treats integration and exclusion as constitutive (Tononi et al., 2016), global-workspace accounts tie availability to broadcast (Baars, 1997; Dehaene et al., 2014), and predictive-processing accounts derive both from a hierarchical generative model (Friston, 2010). Here the three requirements are kept apart rather than derived from one another, so that each carries its own assumptions and admits its own measurement; Section 9.4 returns to the comparison.

Consider reaching for a cup. A visual description of its location and a description of the hand’s relation to it can agree about where the cup is while contributing different contents. Synchronizing the two processes would leave disagreement about that shared quantity unresolved. An internal representation must additionally locate the perceiver within the represented situation. This example motivates the requirements; the mathematical constructions use spatial measures and fields in real normed vector spaces, whose identification with perceptual content is a further modelling task.

The mathematical contribution is to specify connections between these requirements and the assumptions each connection needs. A declared actuator’s stored energy bounds its mean coupling. Phase encoders shared on

overlaps convert phase order into a bound on content disagreement. Compatible spatial measures have a unique global extension; approximate agreement controls a specified global selection. A contracting reconstruction makes that extension its unique fixed point only if it accurately reconstructs the extension. Accuracy across a family of states imposes a further condition on the information retained by the encoding.

The empirical contribution is to identify what observations can decide. Stationary phase distributions can be shared by stronger coupling, weaker diffusion and a matched common drive. Finite ramps change the observable onset. A winding field is stationary at the same coupling as a uniform one and has the opposite resultant, so that observable does not determine the state. Coherent adaptation can lose alignment with a designated environmental template, and a compatibility estimator needs diagnostics that distinguish content from decoder artifacts. These findings determine the measurements needed to test the proposed connections.

Three of the ingredients are standard: the synchronization threshold $K_c = 2D$ for the noisy mean-field model (Sakaguchi, 1988), the unique gluing of compatible local sections, and the Banach fixed-point theorem (Banach, 1922). Four things here are not. The installed-energy condition bounds mean coupling by the stored energy of a declared actuator, which makes a resource condition necessary for coherence on the stationary branch. The coherence-to-content bound converts phase order into a statement about descriptions rather than about timing, and its patch-local form is what keeps

58 it from being vacuous in a large population. A stability criterion on the
59 reweighted coupling kernel, together with an existence condition on an am-
60 plitude layer, settles when a low resultant reports weak coupling and when
61 it reports a winding field. And the composition is machine-checked together
62 with a witness discharging all eight hypotheses at once, which is what makes
63 their conjunction non-empty rather than a list no object satisfies.

64 Granting those premises, collected in Table 1, the account runs as follows.
65 Extracellular contraction on awakening raises endogenous field amplitude
66 and with it the effective coupling between cortical oscillators. Once that
67 coupling passes $2D$, a coherent phase branch exists, and from an initial spread
68 within a quarter turn the population settles onto it; other initial data leave
69 windings stationary at the same coupling. Encoders shared where cortical
70 territories overlap turn that phase order into agreement about the quantities
71 the territories describe in common, so the local descriptions glue: there is
72 one state of which each is a restriction. Part of that state encodes the state
73 itself, and a readout of that part reconstructs it accurately enough, and
74 sensitively enough across the states that matter, to make it the unique state
75 the system models itself as being. On this account the glued state is what
76 makes distributed activity one situation, and its accurate self-reconstruction
77 is what gives that situation a perspective. Every step of it is conditional.

78 Section 2 collects the results that hold whatever physical field is proposed
79 and whatever substrate carries it; a reader who rejects the cortical hypothesis
80 of Section 8 still has to accommodate them. Sections 3–5 then develop the

81 three requirements together (Fig. 1). Section 6 composes them, stating the
 82 thermodynamic constraints and all eight connecting hypotheses per branch;
 83 its supporting models of feedback, learning and resource supply have distinct
 84 state spaces and assumptions. Section 7 then examines what the observables
 85 can establish, before Section 8 develops the cortical hypothesis and an awak-
 86 ening experiment. The hypothesis is that extracellular contraction increases
 87 field-mediated coupling; its direction, rate and relation to recovery require
 88 independent calibration. Supplemental Material organizes the constructions
 89 and controls behind the claims they support.

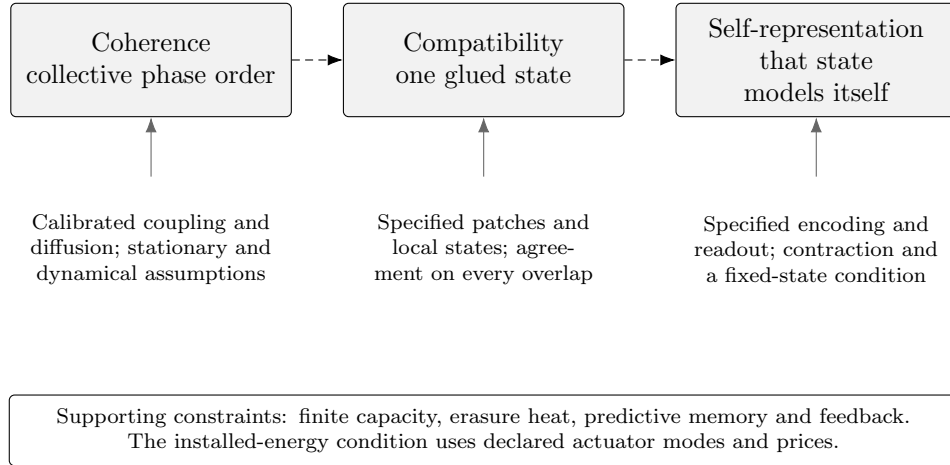


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

90 2 What holds without the cortical hypothesis

91 Most of what follows is conditional. Table 1 names eight hypotheses that
92 the composition consumes rather than derives, and the cortical identification
93 of Section 8 is a further commitment on top of them. Five results are not
94 conditional in that way. They constrain any account that reads unity off
95 phase coherence, whatever physical field is proposed to carry the coupling
96 and whatever substrate carries the field. Each is stated below where it is
97 used; collected here, they are the article’s fixed points.

98 **Phase order alone does not constrain content.** Assign every patch
99 the same phase and let two local mass profiles disagree at a shared site.
100 The profiles are chosen independently of phase, so synchrony bounds their
101 disagreement not at all, and they have no common extension: perfect locking
102 is consistent with local descriptions that do not glue (Section 4). What the
103 positive result of Section 4.2 costs is exactly this gap. A bound on content
104 disagreement needs encoders already related on overlaps, and that relation
105 is a supplied fact about the descriptions rather than a consequence of the
106 dynamics.

107 **The order parameter does not determine the state.** A field winding
108 once across a periodic sheet has $r = 0$ while every patch of it stays locked,
109 and this is not a contrived initial condition. On a periodic array of any
110 dimension, a winding is stationary for *any* isotropic symmetric kernel at

111 identical frequencies, because the coupling term is odd in the separation
 112 against a kernel even in it (`char_is_kuramoto_trajectory`). Section 7.3
 113 exhibits a sheet on which uniform and winding initial conditions reach 0.9657
 114 and 0.0083 at one coupling, diffusion and kernel. A low resultant therefore
 115 reports either weak coupling or a winding field, and no measurement in r
 116 alone separates them. The state also carries an integer the resultant does
 117 not: read around a ring, the same winding has winding number q , and a
 118 nonzero q obstructs every global real-valued phase lift (`winding_degree_`
 119 `obstructs`). The resultant is an average over sites and moves when they are
 120 reweighted or the patches redrawn; the degree telescopes and does not.

121 **A unique fixed point is not a self-model.** A constant readout con-
 122 tracts, fixes whatever state it returns, and reconstructs nothing. Uniqueness
 123 is consequently the wrong criterion, and Section 5.2 gives the right shape
 124 of one: reconstructing M states pairwise more than 2ε apart within error ε
 125 requires M distinguishable codes, so an alphabet of 2^b codes caps the family
 126 at $M \leq 2^b$. This is a counting argument and holds for any encoder and
 127 readout whatever.

128 **Where the encoding is a restriction, the region must resolve the**
 129 **family.** Read the code as the state's own restriction to a region and
 130 the criterion moves off the readout entirely. A boundary whose region
 131 reads two relevant states identically cannot reconstruct them more than 2ε
 132 apart (`not_isRestrictionResonance_of_reconstructs`), and a resonant

133 boundary that does reconstruct a separated family restricts injectively on
134 it (`restrictToAvatar_injOn_of_isRestrictionResonance`). Where the
135 region sits is fixed before an encoding is chosen, so no readout supplies
136 resolution the region lacks.

137 **Agreement after a change takes communication.** In a synchronous
138 network whose updates read only the site’s own state and its declared neigh-
139 bours’ messages, two configurations agreeing on a site’s causal past through
140 T rounds leave that site identical at round T . Two interventions outside
141 that past that require reports more than 2ε apart therefore cannot both be
142 reported from it within ε by the deadline (Section 5). Snapshot compatibil-
143 ity and timely compatibility are therefore different requirements, and cutting
144 the path removes the second while leaving the first intact.

145 None of these depends on the coupling being electromagnetic, on the
146 substrate being cortex, or on the eight hypotheses of Table 1 holding. Sec-
147 tion 9.3 applies the last three to a candidate that is neither biological nor
148 field-coupled.

149 3 Coherence: physical coupling and its re- 150 source condition

151 3.1 Dynamics and the stationary threshold

152 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

153 Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and
154 K_{ij} has inverse-time units, with angles dimensionless. Uniform mean-field
155 coupling is $K_{ij} = K/N$; frequency heterogeneity and diffusion are distinct
156 parameters. The stationary calculation below uses identical frequencies and
157 this scalar coupling. Applying it to a spatial kernel requires an aggregation
158 rule and a check of the resulting approximation.

159 For identical frequencies in their rotating frame, the positive normalized
160 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

161 where I_n are modified Bessel functions and ψ is the collective phase. The for-
162 mal derivation uses the absence of stationary current for a periodic gradient
163 drift to obtain this density, then verifies normalization and self-consistency.
164 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988). Above

threshold, exactly one positive solution coexists with zero and increases
strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

Figure 2 shows this stationary branch.

Linearization about the uniform density gives rate $K/2 - D$ for the first harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists exactly above threshold. About the coherent density, the quadratic form is strictly dissipative transverse to collective rotation, whose rate is zero. These linear estimates concern the limiting equation. Relating them to finite stochastic trajectories requires a dynamical mean-field limit and branch selection; exchange symmetry alone does not establish the needed independence (Sznitman, 1991).

3.2 Installed energy bounds mean coupling

A spatial model replaces the finite sum by integration against a kernel. One declared implementation carries response modes with continuous profiles ϕ_i , installation prices u_i and nonnegative occupancies $c_i(z)$ in configuration z :

$$K_z(x, y) = \sum_i c_i(z) \phi_i(x) \phi_i(y), \quad U(z) = \sum_i u_i c_i(z). \quad (4)$$

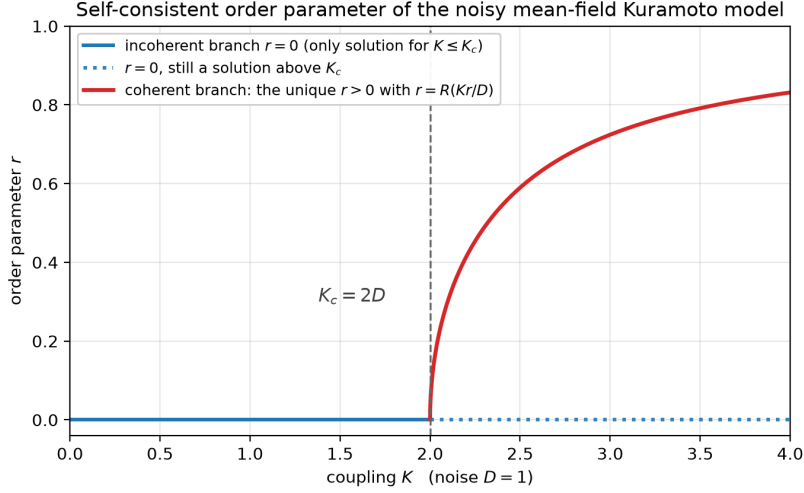


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

180 The same configuration sets the reciprocal kernel and stored energy. An
 181 executed transition changes the kernel through its changed occupancies and
 182 requires work $U(z') - U(z) + Q$, with Q positive into the reservoir. The
 183 installed kernel is averaged over the resulting configuration law $p(z)$. Cell-
 184 pair reconstruction on refining spatial partitions converges to its product-
 185 measure integral under the specified compactness and sampling assumptions;
 186 the refinement index is independent of the number of control updates.

187 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 188 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's

189 integral gives

$$\begin{aligned}
 U_{\text{inst}} &= \sum_z p(z) U(z), \\
 K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_z p(z) \sum_i c_i(z) m_i^2 \leq \kappa U_{\text{inst}}.
 \end{aligned} \tag{5}$$

190 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 191 coherent stationary branch and any growing Fourier mode about the uniform
 192 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 193 is necessary: energy above it need not produce sufficient coupling because
 194 the bound can be loose.

195 The quantity constrained is installed energy rather than dissipated heat,
 196 and how much the condition excludes depends entirely on κ , which the mode
 197 profiles, prices and substrate normalization supply. Any macroscopic sub-
 198 strate carries installed energy far above the thermal floor, so the inequality
 199 binds through that conversion and not through the energy, and no result here
 200 derives κ for cortex or for any other candidate; what the condition contributes
 201 is a resource requirement in units a measurement can reach. Applying the
 202 scalar threshold to heterogeneous or signed spatial dynamics requires a jus-
 203 tified reduction, and identifying the installed kernel with a cortical field is a
 204 separate physical commitment.

205 4 Compatibility: from coordinated activity 206 to a global description

207 4.1 Why common timing is insufficient

208 For overlapping patches U and V , compatibility means that restricting either
209 local description to $U \cap V$ gives the same object. Descriptions can agree on
210 shared quantities while contributing different information elsewhere. Com-
211 patible sections of the specified sheaf have a unique global extension. On
212 a finite cover, local finite spatial measures agreeing as measures on every
213 overlap also have a unique finite-measure extension whose sheaf image is the
214 glued section. No mass-one normalization is imposed.

215 In the cup example, visual and reaching descriptions must express a
216 shared quantity, such as cup location, in common coordinates. Restriction
217 retains that quantity while dropping details particular to each description.
218 Agreement permits surface appearance and hand configuration to contribute
219 to one situation. The mathematical counterexample assigns the same phase
220 to every patch while two local mass profiles disagree on a shared site. Those
221 profiles have no common extension. Because they are chosen independently
222 of phase, synchrony places no constraint on their disagreement.

223 This distinction limits the biological interpretation. Receptive territories
224 and content domains must be related to spatial patches, and the descriptions
225 must support the required extension. Matching overlap marginals of de-

226 coded probability laws need not determine a unique global joint distribution
 227 (Abramsky & Brandenburger, 2011).

228 The cover is itself a choice, and the construction constrains it in one way
 229 only: synchronized patches agree where they overlap. Where they do not
 230 overlap there is nothing to compare, so a cover by pairwise disjoint patches
 231 satisfies that obligation for *every* assignment of local descriptions and phases
 232 (`ofDisjointCover`) and the gluing theorem still returns a unique global sec-
 233 tion, which is the tuple it was handed (`glue_unique_of_disjoint`). Member-
 234 ship of the class then reports nothing about the system beyond the patches
 235 covering it. What a gluing claim asks of a cover is therefore a non-degeneracy
 236 the theorem does not assume: that distinct patches meet (`HasNonempty-`
 237 `Overlaps`), and that what they share is a quantity fixed independently of
 238 the agreement it is being used to establish.

239 4.2 When coherence does constrain content

240 A declared relation between phase and content gives a positive result. Let
 241 contents take values in a real normed vector space E . Patch i reads an
 242 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 243 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 244 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 245 sites; their equality on overlaps is a supplied relation between descriptions of
 246 the shared quantity. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$
 247 implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every

248 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1-r^2}. \quad (6)$$

249 Here discrepancy is the norm of the difference, and r and N are read on
250 whichever population the bound is applied to. That choice is what decides
251 whether the estimate says anything. A patch carries an order parameter of
252 its own, and reading the bound there carries that patch's resultant and that
253 patch's site count (`chord_le_of_patch_coherence`, `compatible_of_patch_`
254 `coherence`), so a large population observed through small patches is con-
255 strained by the small count. Read instead on the whole array, the factor N
256 makes the estimate weak, because it draws a pointwise guarantee from an
257 average over N^2 pairs. At $r = 1$ the observations agree exactly at either
258 scale. A shared scalar encoder is a special case. Dropping the overlap rela-
259 tion permits disagreement even at perfect locking. A common nonexpansive
260 linear projection preserves agreement, but agreement of a projection does
261 not establish agreement of the full vector.

262 What Eq. (6) bounds is a chord, which never exceeds 2, so the estimate
263 constrains anything only where $\sqrt{2} N\sqrt{1-r^2}$ falls below that. The winding
264 sweep of Section 7.3 fixes both ends of that range on one sheet. Read on the
265 sweep's own patch, 625 sites at patch-local order 0.9015, the bound is $382.5 L$
266 and constrains nothing; holding that order fixed, the count alone brings it
267 below the maximum chord only at 3 sites or fewer. A nearest-neighbour pair

268 of the same winding, the smallest patch there is, gives $0.1388 L$. Shrinking
 269 patches therefore thins the overlaps at the same rate as it tightens the bound,
 270 and the two ends are not reached by the same cover. At a columnar grain
 271 the estimate constrains agreement inside one locally synchronous population
 272 and is vacuous across the distant territories whose agreement unity is about.

273 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 274 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 275 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 276 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

277 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 278 duce an initial mismatch. Approximate observations leave a nondecaying
 279 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 280 update to the phase process.

281 4.3 What approximate agreement determines

282 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
 283 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
 284 Weights that vanish outside their patches and sum to one at each site select
 285 a global profile by weighted averaging. The selection lies within ε of each
 286 patch there, and any two such selections differ by at most ε , independently
 287 of cover multiplicity. An L -Lipschitz readout consequently varies by at most

288 $L\varepsilon$ across selections.

289 At zero tolerance, the selection is the unique exact extension. At every
290 positive tolerance there are families with no exact extension; selection then
291 gives an approximation whose dependence on weights is bounded. Supple-
292 mental Material gives a three-site witness attaining the bound. The mass-
293 profile construction uses the scalar case of the content dynamics and addi-
294 tionally requires nonnegativity of the initial profiles and observations. The
295 general vector result supplies no identification with the spatial-measure sheaf.

296 These constructions concern declared objects: a specified cover, contents
297 valued in a normed space, and encoders related on overlaps. They select
298 neither the neural cover nor the decoder. Identifying their contents with
299 perceptual content, and establishing that the identification bears on experi-
300 ence, is a further modelling task; Section 7 sets out what a measurement of
301 it would have to establish.

302 **5 Self-representation: a state that models it-** 303 **self**

304 **5.1 The reflexive construction**

305 Compatibility gives a global state. Reflexivity relates it to an encoding within
306 that state. Equip global sections with a complete metric. If the additional
307 restriction-resonance condition identifies encoding with restriction to a region

308 U , the reconstruction map is

$$F(s) = \text{readout}(s|_U). \quad (8)$$

309 The region and readout are fixed across the state space. If F contracts and
 310 fixes the section s_* formed from the local descriptions, Banach's theorem
 311 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89
 312 supplies the contraction and $F(s_*) = s_*$; restriction resonance is separate.
 313 Supercritical coupling makes the stipulated factor $q = e^{-(K-2D)\tau/2}$ less than
 314 one for $\tau > 0$. Constructing a map with that factor remains a modelling
 315 obligation.

316 In the cup example, the global state includes the represented scene and the
 317 perceiver's relation to it. A local encoding must retain enough information
 318 for the readout to recover that modelled state; cup position alone would be
 319 insufficient if hand configuration and viewpoint could not be reconstructed.

320 For the same map with Lipschitz factor $q < 1$, approximate reconstruction
 321 gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - q}. \quad (9)$$

322 At the stipulated rate this radius grows as $2\varepsilon/((K-2D)\tau)$ near threshold, for
 323 fixed positive ε and τ . Exact uniqueness survives while the error guarantee
 324 weakens.

325 5.2 Reconstruction across a family of states

326 Sensitivity can be required quantitatively. Declare relevant macrostates, an
 327 encoder E and readout R , and require $d(s, R(E(s))) \leq \varepsilon$ throughout that
 328 family. If two states have the same code, the triangle inequality gives

$$E(s) = E(t) \implies \max\{d(s, R(E(s))), d(t, R(E(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

329 Thus M relevant states pairwise more than 2ε apart require at least M
 330 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
 331 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 332 existing self-prediction map.

333 This is an additional accuracy criterion on a family, with no requirement
 334 for multiple exact fixed points. A blind encoder or constant readout fails it
 335 for separated states despite having a unique fixed point.

336 Where the encoding is read as the state's own restriction to a region, the
 337 criterion becomes a condition on the region. Restriction resonance requires
 338 the encoder and that restriction to identify the same pairs of states, so a
 339 boundary reconstructing two relevant states more than 2ε apart while its re-
 340 gion reads them identically is not resonant (`not_isRestrictionResonance_`
 341 `of_reconstructs`); read forwards, a resonant boundary that reconstructs
 342 a pairwise separated family restricts injectively on it (`restrictToAvatar_`
 343 `injOn_of_isRestrictionResonance`). Which states a region reads identi-
 344 cally is fixed by where the region sits, before an encoding is chosen, so the

345 resolution the family demands is the region's and no readout supplies it. The
 346 implication runs one way: a region that does resolve the family is not thereby
 347 shown to carry a resonant encoding, and E89 supplies no such resolution.

348 Those two demands are jointly satisfiable with nothing folded anywhere.
 349 Take the region to be the whole substrate: restriction is then the identity, so
 350 the boundary is resonant, reconstructs every declarable family exactly, and
 351 restricts injectively on all of them (`topAvatar_satisfies_obligations`). A
 352 properness condition on the region (`IsProperAvatar`) is therefore a separate
 353 requirement rather than a consequence of resonance and resolution, and what
 354 a fold into a local region is asked for. Its quantitative form replaces the
 355 alphabet by a metric. With an L -Lipschitz readout, a family reconstructed
 356 within ε whose members lie pairwise more than r apart has δ -separated codes
 357 whenever $\delta L \leq r - 2\varepsilon$, so the family is no larger than the δ -packing number
 358 of what the region carries (`encard_le_packingNumber_restrictToAvatar`).
 359 That is the shape a continuous family of states needs, and at the whole
 360 substrate it says nothing, because the region's range is everything. The gain
 361 L is declared and derived here for no physical readout; an unbounded-gain
 362 decoder is subject to no constraint from this bound.

363 5.3 Agreement after a change takes communication

364 Snapshot compatibility does not specify how quickly descriptions respond
 365 when a shared quantity changes. In a synchronous network whose updates
 366 read only the site's own state and declared neighbours' messages, two initial

367 configurations agreeing on a site’s causal past through T rounds produce the
368 same state there at round T . If two fresh interventions outside that past
369 require reports separated by more than 2ε , no report from that site can be
370 within ε of both. This follows from the reconstruction bound in Section 5.2.

371 A three-site delay line witnesses failure after one round and exact report-
372 ing after two; cutting the path prevents delivery at every round. A report
373 may still happen to be correct about one fixed world. The bound concerns
374 responsiveness across interventions.

375 Throughout this section *prediction* names a reconstruction map; reading
376 it as temporal neural prediction would require dynamics. The relevant family
377 of states, the metric, the tolerance, the communication graph and the dead-
378 line are all declared inputs, as are the represented variables and the fidelity
379 counted as accurate. Reading the resulting code count as physical memory
380 capacity requires a hardware model, and a readout that varies with the state
381 requires a further one.

382 6 Thermodynamics and composition

383 The phase and representation results concern coupling, content disagreement
384 and reconstruction error. Thermodynamics constrains the physical opera-
385 tions that could implement them, and it does so from the side rather than
386 along the spine: no edge below derives coherence, compatibility or reconstruc-
387 tion from a heat bound, and the resource condition these edges terminate in

388 is necessary without being restrictive for any macroscopic candidate (Sec-
 389 tion 3.2). What this half contributes is a ledger in which a resource claim
 390 is stated in checkable units. Finite-register capacity obeys $H(p) \leq \log |X|$;
 391 erasure heat depends on specified system–bath assumptions (Landauer, 1961;
 392 Norton, 2005; Shenker, 2000). A separate boundary result concerns a field
 393 reaching disconnected vacuum components (Kibble, 1976). The composition
 394 connects these objects through the explicit hypotheses in Table 1.

395 **6.1 Feedback changes the information constraint**

396 For passive memory X_t driven by an environmental signal S_t , the predictive
 397 bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (11)$$

398 The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still
 399 et al., 2012). Zero retained information also satisfies it, so a task and a
 400 learning rule require additional specification. Relating this bound to Lan-
 401 dauer heat requires a common thermal scale and budget, and memory waste
 402 bounded by erased entropy.

403 An acting agent changes the signal it will receive. Specify policy, world,
 404 observation and memory-update channels. For their finite laws,

$$\begin{aligned}
 I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\
 &\quad - I(X_t; S_{t+1} \mid S_t).
 \end{aligned} \quad (12)$$

405 The second conditional term vanishes in the passive model. With feedback,
 406 the signed difference can be negative as action creates correlations with the
 407 future. A causal reading requires an environmental state sufficient to describe
 408 the relevant dynamics.

409 For a controlled substep, the active thermodynamic bound instead uses
 410 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \| P_R) =$
 411 $\Delta H(X, S) + \langle Q \rangle / \theta \geq 0$ under local detailed balance, with $\theta = k_B T > 0$ and
 412 heat positive into the reservoir (Horowitz and Esposito, 2014; Ito and Sagawa,
 413 2013). Every forward path of positive probability must have a positive-
 414 probability reverse for this finite identity. A missing reverse gives infinite
 415 divergence. The supplement develops the support conditions and composes
 416 actuation with memory updates on their shared intermediate law.

417 **6.2 Learning and repeated operation need explicit re-** 418 **sources**

419 The finite examples distinguish selecting a feasible policy, relaxing a register
 420 toward an encoded reward, and learning an unknown task from observa-
 421 tions. The observational learner acquires correlation with an environmental
 422 parameter while its action marginal stays uniform. A continuing model exe-
 423 cutes actuation, observation-driven learning and reset in sequence, preserving
 424 above-chance performance at a positive work cost per cycle.

425 A phase-derived sensor makes the informational distinction concrete: an

426 informative readout and a blind control have the same phase order but dif-
 427 ferent learning performance. Attaching the learner’s register to the actuator
 428 of Section 3.2 ties reward, installed kernel and work to its own executed law.
 429 The learning witness rises above the necessary installed-energy bound while
 430 its matched control remains on the exclusion line. These finite constructions,
 431 detailed in Supplemental Material, leave the evolving feedback between phase
 432 dynamics, installation and sensing to be specified.

433 For that model, common energy E , thermal scale θ , local detailed balance
 434 and positive support at each stage give

$$\theta[H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle E \rangle_0 - \langle E \rangle_N \leq B + \langle E \rangle_0 - \langle E \rangle_N, \quad (13)$$

435 where expected cumulative work stays within the supplied allowance B at
 436 every prefix. This bounds expectations. Separate constructions track a store
 437 on each trajectory, include finite sources, charge sensor clearing and prepend
 438 preparation. They specify when a mean allowance represents an available
 439 resource; an expected budget alone permits overdrawn paths.

440 These constructions establish costs for their declared processes. A finite
 441 source with a positive net cost permits only finitely many funded cycles,
 442 while continuing operation requires replenishment. Register–bath witnesses
 443 also distinguish actual bath-energy transfer from transition-log heat. Iden-
 444 tifying a work store, an erasure budget and the installation energy of Sec-
 445 tion 3.2 requires physical relations that these separate models do not supply

446 automatically.

447 **6.3 What the composition establishes**

448 The passive theorem `chain` and feedback theorem `chain_active` each ac-
 449 cept eight connecting implications (Table 1). In the active branch, a named
 450 substep at the register’s thermal scale satisfies

$$\theta > 0, \quad \theta[H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (14)$$

451 with B the register’s heat. The upper allocation follows from a ledger whose
 452 operations deliver that total heat to one reservoir and whose other operations
 453 do not increase joint entropy. Those ledger assumptions are physical inputs.
 454 Neither Landauer’s lower bound nor microscopic reversibility supplies them.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12 <i>capacity and boundary</i>	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.
E23 <i>boundary to register</i>	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.

Table 1 (continued)

Connection	Status	Required assumption
E34 / E34Active <i>register to dissipation</i>	Bridge assumption	Passive: predictive memory at the register's thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations' compressiveness.
E45 / E45Active <i>couplings to continuum</i>	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Refinement and physical identifications remain inputs.
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours.
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $q < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.

Table 1 (continued)

Connection	Status	Required assumption
Installed energy	Conditional theorem	Under Section 3.2’s occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

Both branches also take a thermodynamic cover carrying exact overlap compatibility and a reflexive boundary specifying encoding and readout. They conclude **UnifiedSelf**: the glued section is the unique fixed point of self-prediction. Compatibility and accuracy at that section enter as premises; uniqueness follows. Identifying encoding with restriction needs the separate resonance condition of Eq. (8).

Each of the eight edges is supplied rather than derived from the one above it. E12 is equivalent to its conclusion, so capacity and boundary are independent roots rather than a step from one to the other; both E45 discharges take their convergence from refinement data and not from the thermodynamic bound; E78 states that coherence relaxes the supplied cover, and the coherence-to-content estimate of Eq. (6) bounds disagreement only for encoders already shared on overlaps. The inferential content of the composition is its last step, where contraction and a fixed glued section give uniqueness. What composing them establishes is that the premises are consistent, named in one place and machine-checked together, and not that coherence follows from installed energy or self-representation from coherence.

Joint witnesses combine a register, boundary, refining mesh and small

473 spatial cover; active witnesses add controlled thermal updates and a priced
474 installation. They establish satisfiability under their stated identifications.
475 Mesh convergence follows from refinement data, and the witnesses establish
476 no common cortical mechanism. Supplemental Material maps the claims to
477 their Lean declarations and records the counterexamples. The development
478 declares no additional axioms beyond its libraries' foundational axioms.

479 **7 What the observations establish**

480 The composition holds under its premises. Whether observation can estab-
481 lish those premises is a separate question, and the three requirements it
482 connects concern different objects: phase dynamics, local descriptions and
483 internal encodings. Their observables must retain these distinctions. The
484 following controls show which mechanisms remain indistinguishable, which
485 temporal approximations fail, which states one observable cannot separate,
486 and what independent content measurements must establish. Each is a finite
487 numerical regime on a declared model, with full parameters and controls in
488 Supplemental Material, and none of them calibrates a cortical quantity; the
489 scope notes below concern what each result does and does not separate.

490 **7.1 Stationary distributions leave the mechanism un-** 491 **determined**

492 Equation (2) gives a concentration–coherence relation with no fitted shape
493 parameter. Estimate r from circular resultant length and a from density
494 shape: inverting R from the measured r would make agreement automatic.
495 Even independently estimated agreement identifies a statistical family shared
496 by different mechanisms.

497 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
498 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
499 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
500 oscillators with prescribed drift $-h(t) \sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
501 and diffusion D_0 have that same density. Their stationary distributions co-
502 incide window by window despite different causes.

503 Resolved dynamics supply additional information. In the ideal model with
504 continuous paths, quadratic variation accumulates at rate $2D$ and conditional
505 increment means constrain drift. Separating endogenous coupling from a
506 matched common drive additionally requires a measured drive or controlled
507 perturbation; identifying electromagnetic coupling requires evidence specific
508 to that interaction. Supplemental Material gives the equivalence construction
509 and its numerical check.

510 **7.2 Finite ramps change the observable onset**

511 The square-root law in Eq. (3) describes stationary solutions. A changing
512 coupling requires time for the phase distribution to approach those solutions.
513 In a deterministic mean-field control that crosses the threshold symmetrically,
514 none of the scanned rates meets the declared tracking tolerance: slowing
515 the crossing does not remove the terminal residual under that initialization.
516 A slow control variable therefore supplies no general guarantee of branch
517 tracking at onset. Supplemental Material gives the residual, rate range and
518 admissible rates away from threshold.

519 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
520 at 8 coupling speeds from $v = 0.1$ to $v = 0.0001$. Escape is the first of three
521 consecutive saved samples at $r \geq 0.2$ after threshold. Faster ramps delay
522 escape (Fig. 3); only 5/32 replicas escape within the fastest window. The
523 6 uncensored legs supply the delay scaling below; the decade-spaced ones
524 among them have mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted
525 onset exponents differ from those of the exact stationary branch evaluated
526 on the same grids. At the fastest speed the ensemble mean never reaches the
527 fit window, so its fitted slope does not estimate stationary onset.

528 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
529 makes the apparent exponent depend on ramp speed and fitting window.
530 The specified deviation-based critical ramp speed is not bracketed by this
531 sweep. With fluctuating coupling, the relevant stationary comparison also
532 depends on rate: the mean coupling approximates order in the tested fast

533 regime, while averaging stationary order over coupling works for the tested
 534 slow drives confined to one side of threshold. Slow crossing drives can satisfy
 535 neither approximation.

536 **The prediction.** The delay scaling is what this section supports as a pre-
 537 diction, and it needs no absolute calibration. Over the 6 uncensored legs the
 538 mean delay follows $\Delta K \propto v^{0.444}$, with a 95% slope interval of $[0.394, 0.494]$.
 539 The reference for that exponent is the same estimator applied to a deter-
 540 ministic mean-field run seeded at the fluctuation floor, which returns 0.425
 541 over the same legs at the same escape criterion, inside the interval on the
 542 measurement. The distance from $1/2$ is what an absolute escape criterion
 543 costs: the run crosses the saturation between the floor it starts from and the
 544 level it is stopped at, and tightening the criterion to $r \geq 0.05$ returns 0.490.
 545 Stated as a law rather than as a number: onset delay scales as the square
 546 root of the rate at which coupling crosses threshold. Testing it requires only
 547 that $\dot{K}$ be monotonic in a manipulated rate — not its value, nor κ , nor γ
 548 — so it is a within-subject design that varies the rate of emergence from
 549 anaesthesia alone, with each participant as their own control. Such a design
 550 must avoid the hazard this sweep exhibits: a fixed absolute escape criterion
 551 meets a size-dependent critical fluctuation floor, so the criterion has to be
 552 referred to each recording’s own pre-threshold fluctuation scale rather than
 553 to a shared number. The diffusion D is the remaining input, and Section 7.1
 554 notes that it is estimable from resolved phase traces through the rate $2D$

555 at which quadratic variation accumulates, which is a different quantity from
 556 the concentration the propofol exercise could not range over.

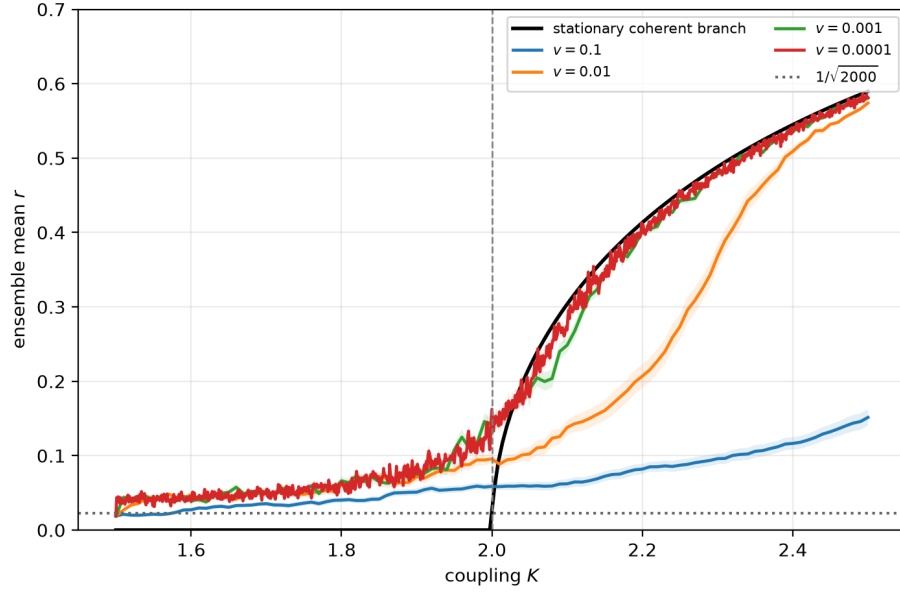


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

557 Additional controls address integration bias, population size and spatial
 558 or signed coupling; the refined signed-network crossing bracket $K_{\text{eff}}/D \in$
 559 $[1.958, 2.009]$ contains 2.

560 **7.3 A winding state leaves the order parameter am-** 561 **biguous**

562 The order parameter r is indifferent to where a phase sits in space. A field
563 winding once across a periodic sheet therefore has $r = 0$ while every patch
564 of it stays locked. Such a field is not an artificial construction here. On a
565 torus a winding is stationary for any isotropic symmetric kernel at identical
566 frequencies: the coupling term is odd in the separation against a kernel even
567 in it, so the drift cancels pair by pair. `char_is_kuramoto_trajectory` proves
568 this for any finite periodic array, this sheet included. A winding satisfies the
569 positivity and identical-frequency hypotheses. It violates only the quarter-
570 turn initial spread that the convergence result additionally requires before
571 placing the limit at the uniform state.

572 On the sheet of Section 3's spatial control, a winding survives below
573 an interpolated decay length of 0.1871 mm and unwinds above it. That
574 boundary is interior to the 0.1–0.3 mm band. At 0.1 mm the same sheet at the
575 same coupling, diffusion and kernel reaches $r = 0.9657$ from a uniform start
576 and $r = 0.0083$ from a winding one. Both are stationary, with patch-local
577 order 0.9015 and no winding defect. At fixed K the observable therefore takes
578 two values spanning nearly its whole range. Coupling does not determine the
579 resultant, and a low resultant does not by itself distinguish weak coupling
580 from a winding field.

581 Under the quenched frequency spread of 1.5 rad s^{-1} that the spatial con-

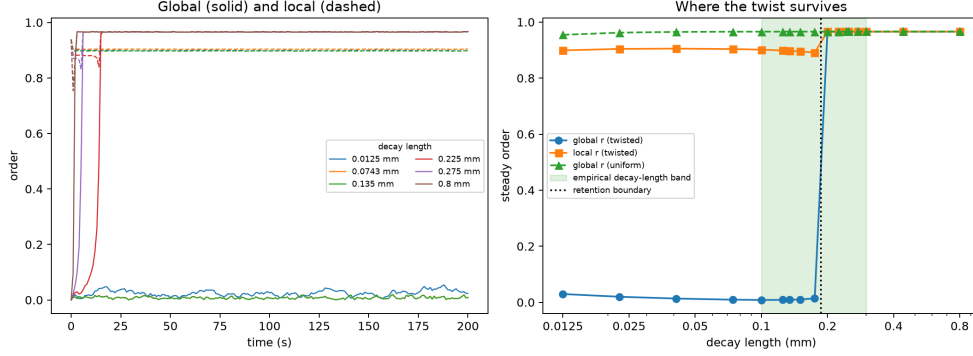


Figure 4: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

582 trol carries, the boundary moves to 0.1421 mm. The same comparison at
 583 0.1 mm gives 0.9418 against 0.0348, holding across 3 frozen realizations.
 584 Heterogeneity shortens the range that sustains a wave without removing it.

585 Which windings persist has a criterion behind it. Linearizing the phase
 586 model about a winding returns a coupling of the same translation-invariant
 587 form, each separation reweighted by the cosine of the phase the winding
 588 advances across it (`hasDerivAt_charJacobian`). That reweighted kernel is
 589 again even in the separation, so the same characters diagonalize it and ev-
 590 ery mode’s rate is one of its eigenvalues. Where the winding advances by
 591 at most a quarter turn across every separation the coupling reaches, the

592 reweighted kernel is nonnegative and no rate is positive (`charJacobian_`
 593 `stable_of_quarter_turn`); where some rate is positive, that mode grows
 594 without bound (`charJacobian_growing_mode`). Moving coupling mass to
 595 longer separations is what carries a phase advance past a quarter turn, and
 596 that is the mechanism the decay-length sweep varies. The criterion bounds
 597 rates rather than trajectories: asymptotic stability is not proved, so the
 598 states that persist occupy a narrower range than it admits, and the bound-
 599 ary quoted above is measured and not derived from it.

600 The same reweighted kernel measures the potential's curvature. A wind-
 601 ing has $r = 0$ where a global minimizer of the Lyapunov potential has $r = 1$,
 602 so it is not one (`char_not_potential_min`): the quarter-turn condition on ini-
 603 tial data does not merely decline to cover a winding, it must exclude it. Along
 604 a mode, the potential's second variation is that kernel's quadratic form, so a
 605 growing mode is a direction of falling potential and a winding carrying one
 606 is not a local minimum either (`char_not_local_min_of_charLambda_pos`).

607 An amplitude layer decides which of these states exist, and leaves what
 608 the resultant reports unchanged. A phase state carries no amplitude. Ev-
 609 ery oscillator sits on the unit circle, so a field with an amplitude node has
 610 no representation in these dynamics. The real Stuart–Landau extension
 611 supplies one on the same coupling, and leaves intact the gradient struc-
 612 ture the convergence results rest on. The state becomes $z_i \in \mathbb{C}$ with $\dot{z}_i =$
 613 $(\mu - |z_i|^2)z_i + \sum_j A_{ij}(z_j - z_i)$. Its phase equation is the Kuramoto equation
 614 with each coupling weighted by a ratio of amplitudes (`amplitude_phase_is_`

615 kuramoto). The phase model is this one with the amplitude ratios set to
616 one, rather than a separate object.

617 What the added degree of freedom decides is which windings exist at all.
618 On a periodic array of any dimension, this sheet included, a winding requires
619 $\mu + \lambda > 0$. Here $\lambda = \sum_d f_d(\cos \phi_d - 1)$ is the eigenvalue the kernel supplies,
620 summed over separations, with ϕ_d the phase the winding advances across d
621 (exists_charState_iff; on the ring $\phi_d = 2\pi qd/n$). For the nonnegative
622 kernels used here $\lambda \leq 0$, and moving coupling mass to longer separations
623 lowers it further. A broad enough kernel therefore removes the winding out-
624 right. The sweep above reaches that same boundary by a different route:
625 the phase model retains every winding at every coupling, and loses this one
626 under integration instead. The two layers bound the same regime from op-
627 posite sides. A phase recording does not separate a state the amplitude layer
628 excludes from one the dynamics fails to sustain.

629 These states are windings and not spirals, and on the sheet that is a
630 theorem rather than a shortage of dimensions. A cortical phase singularity
631 is a point at which the amplitude vanishes and around which the phase turns
632 once, read in practice as the circulation around a small plaquette. The states
633 above have equal modulus at every site, so that circulation is exactly zero
634 on every plaquette (plaquetteCirculation_eq_zero_of_winding). Nor can
635 the layer host a single defect. Over a periodic sheet the circulations of *any*
636 phase field sum to zero, so defects arrive in cancelling pairs (no_isolated_
637 defect). The sweep's near-zero measured defect density agrees with the first

638 of these and says nothing beyond it.

639 Two limits bound the layer for a reader of the cortical literature. Relax-
640 ational dynamics has no propagation speed, which is the quantity a cortical
641 wave measurement leads with. And $|z|$ is a normal-form radius, which a mea-
642 sured envelope is not; this work identifies the two nowhere. Supplemental
643 Material gives the sweep, its uniform control, the amplitude layer and the
644 standing wave it admits inside the band, together with the remaining limits
645 of both.

646 A vanishing resultant is the absence of global agreement and not of agree-
647 ment. Eq. (6) read on a nearest-neighbour patch of a winding gives a dis-
648 crepancy bound of $2\sqrt{2}L|\sin\phi|$, with ϕ the phase advanced across the patch,
649 which falls to zero as the same winding is spread over more sites (`chord_le-`
650 `of_char_patch`), while the global form of the same bound is vacuous at $r = 0$.
651 Encoders shared on overlaps therefore still bound content disagreement in-
652 side a patch of a state that no global observable calls coherent. The winding
653 number itself stays unconnected to content. It is, however, a property of the
654 state rather than of the cover read on it. On a ring the whole turns dropped
655 by the winding's affine local lifts are zero at every site but the wrap, where
656 the residue resets, and they sum around the loop to q ; a nonzero q obstructs
657 every global real-valued phase lift, so one state at one coupling has resultant
658 exactly zero and degree q (`winding_degree_obstructs`). Reweighting the
659 sites moves the first quantity and redrawing the loop's intermediate transi-
660 tions leaves the second where it was. The scope is the ring's: this is a loop's

winding and not a spiral, the transitions are those of the affine lifts rather than of an unwrapped measurement, and the integer is identified with no cortical observable.

This qualifies what a phase trace can establish. An onset criterion stated in r alone registers a change of basin as readily as a change of coupling. The protocol of Section 8.3 therefore needs a spatial observable beside the resultant.

7.4 Coherent adaptation can lose environmental alignment

An adaptive coupling rule might supply representational structure, but its objective must reward the relevant agreement. For symmetric coupling, pairwise sine terms cancel in the population mean drift. Minimizing squared drift at that fixed mean therefore rewards equalizing drift across nodes. In the frequency-cluster control, this favors coupling between unlike-frequency clusters, whereas the diagnostic environmental template rewards coupling within clusters.

The formalization proves descent of the squared-drift objective for a specified gradient flow at fixed phases. Its functional differs from thermodynamic entropy production: at a coherent stationary density the probability current vanishes while phase-averaged squared drift is positive. Supplemental Material gives the operator, its gradient and the conditions of the descent result.

682 The finite control co-evolves noisy phases and symmetric, nonnegative
683 couplings at fixed total resource. Its template is a diagnostic covariance of
684 latent signals, never a training input. Across three seeds, second-half order
685 remains between 0.9167 and 0.9222, while the within-cluster to between-
686 cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 5).
687 Random-update, permuted-gradient and frozen-kernel controls are reported
688 in Supplemental Material.

689 The immediate post-update objective reduction disappears when aver-
690 aged over complete update intervals. A prespecified grid of 6 rate–cadence
691 combinations contains no case meeting the 5% sustained-reduction criterion
692 while retaining coherence and losing alignment. The finding is coherent adap-
693 tation away from the specified structure, with no demonstrated sustained cost
694 reduction.

695 Kernel alignment and overlap compatibility are different observables.
696 This control addresses acquisition of its designated template under its par-
697 ticular objective. Learning models can explicitly include data in an objective
698 (Corberi et al., 2026); an account of agreement about the cup must specify
699 the content its learning rule is meant to preserve.

700 **7.5 Content agreement needs independent decoders**

701 A neural test needs independently identified domains, local descriptions and
702 restriction maps. Two decoders should estimate a shared content on over-
703 lapping domains before their agreement is examined. Discrepancy requires

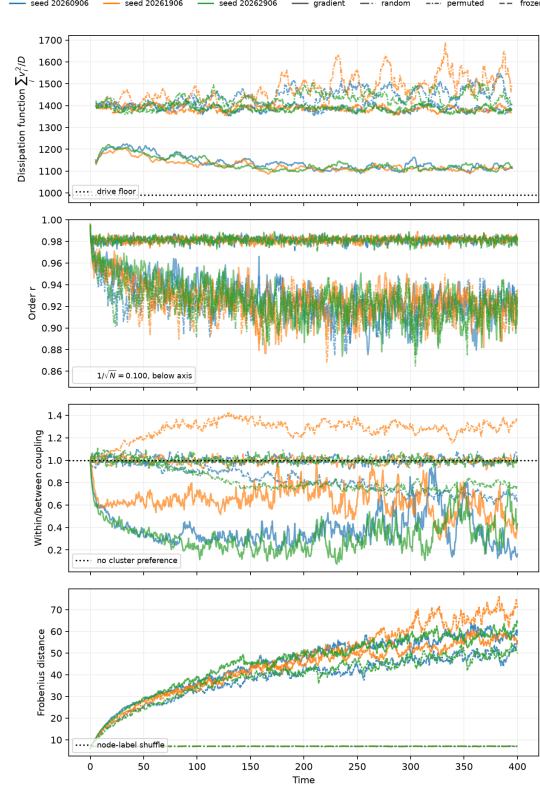


Figure 5: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

704 a null under decoding noise and controls using deliberately compatible and
705 incompatible examples. An implementation on constructed data separates
706 these cases at $AUC = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by
707 trial; the compatible construction scores 0.476 on an interval containing one
708 half. Shared-prior shrinkage can nevertheless manufacture agreement, and
709 regional bias can manufacture disagreement. Per-patch decoding diagnostics
710 detect these failures; the compatibility statistic alone does not.

711 Phase agreement cannot replace that measurement. On constructed con-
712 tent, phase carries a share 1.000 of the content entropy when content is built
713 from it and 0.000 when it is not. A phase-derived statistic reports com-
714 patibility on phase-locked patches whose local descriptions disagree. The
715 implemented content statistic detects their incompatibility. These controls
716 validate a measurement on known constructions; a neural content model and
717 decoder remain to be established.

718 Choosing the cover also chooses a candidate subject: trivial descriptions
719 can agree while carrying little content, and a partition can conceal disagree-
720 ment. The useful choice must therefore be constrained by what neural ac-
721 tivity encodes. The exact composition assumes compatibility of its supplied
722 cover. The content mechanisms above identify additional hypotheses under
723 which agreement can arise; they leave the empirical choice of content and
724 the connection to that cover explicit.

7.6 What an informative phase-density measurement requires

The concentration range and the dependence structure decide whether the concentration–coherence relation can be tested at all. Under the specified log-density estimator and bias-floor criterion, discrimination at the observed concentration ceiling $a = 0.542$ requires 3000 independent sites in the scanned designs, rising to 31000 when sites share a mean in groups of 10. These counts describe that estimator and dependence model; the intended recording requires its own calibration. Supplemental Material reports the bin-count controls and the full dependence-aware design.

That ceiling comes from exercising the estimators on public propofol-sedation EEG (Bajwa et al., 2025), pooling channels in 100-ms windows. Bipolar and common-reference montages produce different residuals. The bipolar results are compatible with the Bessel relation over a concentration range too narrow to distinguish it from its linear approximation at the observed scatter, which is the range the design scan takes as given. Temporal pooling changes the estimand, and sedation recordings leave the awakening hypothesis untested.

An informative dataset therefore needs sufficient concentration range or precision, calibrated estimators and an observation model for dependence, source mixing and reference choice. Public mouse calcium imaging provides thousands of spatially located neurons across waking, sleep and anesthe-

747 sia (Oomoto et al., 2026), but its 7.65 Hz frame rate is below the phase-
748 estimation band used here. Site count, state coverage and bandwidth are
749 separate requirements.

750 **8 The cortical hypothesis and an awakening**

751 **test**

752 The cortical proposal identifies a physical source of coupling and asks whether
753 its changes help explain recovery. The tests in Section 7 determine what
754 observations must distinguish before a phase trace can support this identifi-
755 cation.

756 **8.1 Physical identification and calibration**

757 The cortical hypothesis identifies part of effective coupling with endogenous
758 extracellular electromagnetic interactions. Fields can modulate spike timing
759 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
760 propose coordination of distributed neural information (Pinotsis & Miller,
761 2022; McFadden, 2020). Phase organization under propofol motivates study-
762 ing this regime (Bastos et al., 2021), but does not measure coupling.

763 The formal field predicate requires a jointly continuous kernel, a normal-
764 ized substrate, coupling equal to the kernel average, and no single site pro-
765 viding more than half that strength. Toy constructions satisfy it. A cortical

766 realization must specify the scale at which the kernel and its reduction ap-
767 ply: calcium imaging reports different state-dependent modularity patterns
768 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

769 Cortical columns are the obvious candidate unit, and they answer one
770 of the two demands the formalism makes. As the oscillator index they are
771 a serious proposal: they are delineated by response properties and anatomy
772 rather than by the agreement they would be used to establish (Mountcastle,
773 1997), and a phase can be estimated from a column-sized territory at the
774 electrode spacings used for cortical wave measurements (Townsend et al.,
775 2015). As the content cover they are a poor one, for three separate reasons.
776 A parcellation into non-overlapping columns is a disjoint cover, which satisfies
777 the compatibility obligation without constraining anything (Section 4). The
778 cover condition additionally requires the patches to exhaust the substrate,
779 while much of association cortex has no accepted columnar parcellation. And
780 the column’s standing as a canonical cortical unit is contested (Horton &
781 Adams, 2005; Rakic, 2008). Offering columns for both roles at once therefore
782 supplies the oscillator index and leaves the content cover to be identified on
783 separate evidence, which is the bridge assumption E78 records.

784 The spatial reduction is also testable. A row-sum aggregation at zero
785 diagonal tracks the uniform-coupling control over the tested spatial decay
786 range, while frequency heterogeneity displaces the operational threshold. The
787 fit window biases both estimates relative to $2D$; agreement with the control
788 is therefore the supported conclusion. Supplemental Material reports the

789 reduction, its numerical error and the distinct assumptions needed for the
 790 kernel-average identification in Eq. (5).

791 The field commitment requires physical evidence beyond a choice of math-
 792 ematical representation, and the available support comparisons do not supply
 793 it. A finite matrix spread over positive-mass cells preserves its total cou-
 794 pling and phase correlation exactly, while point-supported vanishing under
 795 an atomless measure and a disadvantage from excluding a better-correlated
 796 pair concern specified support constraints; Section 9.3 reads the same com-
 797 parisons against a digital candidate. Neither supplies energy prices.

798 A spike-time shift alone supplies no coupling rate. The dimensionless
 799 product of population size N , timing sensitivity s , field amplitude E and
 800 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N s E f. \quad (15)$$

801 This rate describes how induced timing shifts accumulate into drift. Nei-
 802 ther it nor the installation factor κ follows from a heat bound. Testing the
 803 physical threshold requires independent estimates of the interaction kernel,
 804 its aggregation and diffusion in the same cortical regime.

805 **8.2 Extracellular geometry and recovery**

806 The proposed control variable is extracellular geometry. Extracellular vol-
 807 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &

808 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
809 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively
810 at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
811 an effective-medium description predicts that a larger extracellular fraction
812 raises conductivity and reduces field amplitude.

813 The coupling-gated hypothesis consequently predicts that extracellular
814 contraction on awakening raises E and, holding the other factors in Eq. (15)
815 fixed, raises effective coupling. This argument supplies a direction; the
816 geometry–coupling relation, its magnitude and its role in human recovery
817 require measurement.

818 For a smooth crossing at nonzero speed, evaluating the stationary branch
819 along the changing control gives the benchmark

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (16)$$

820 This follows from Eq. (3) at fixed D and concerns the stationary branch
821 evaluated at $K(t)$. Reading it as observed recovery requires an indepen-
822 dently justified tracking approximation. The crossing controls in Section 7.2
823 leave that approximation unsupported at onset, even though a separation of
824 geometry and phase time scales can support it away from threshold. The
825 awakening hypothesis therefore needs a calibrated dynamical prediction for
826 an operational onset above the finite-size fluctuation floor. Crossing K_c alone
827 does not supply a recovery latency.

8.3 A protocol for the awakening experiment

The physical test has six requirements:

1. **Coupling and diffusion in the same units.** Estimate them from resolved phase dynamics, separating heterogeneity and observation noise. Measure the drive or perturb the system to distinguish endogenous coupling.
2. **Geometry calibrated to coupling.** Validate the geometry observable and its relation to effective coupling, with uncertainty. The conductivity argument supplies only the expected direction.
3. **Calibrated phase measurements.** Estimate r by resultant length and a by unweighted regression of log density on centered cosine. Choose site count and binning from a dependence-aware calibration over the intended concentration range. The tested designs require thousands of sites at the observed concentrations; temporal pooling changes the spatial estimand.
4. **An independent recovery endpoint.** Measure behavioral recovery separately from phase organization.
5. **A predeclared dynamical onset criterion.** Set a coherence threshold above the finite-size floor and evaluate its latency under the measured coupling trajectory, including ramp speed or fluctuations. Record

848 a spatial observable beside the resultant, since a winding field reports
849 a low resultant at coupling that sustains local order (Section 7.3).

850 **6. Matched model comparison.** Compare coupling increase, diffusion
851 change, common drive and alternative recovery models under the same
852 observation and filtering assumptions.

853 Failure of the concentration–coherence relation over an informative range
854 constrains the density model. Failure of the calibrated geometry–coupling
855 association constrains the physical gate. Better prediction of independent
856 recovery by a competing mechanism counts against its proposed role. A
857 favorable outcome still leaves the content of coordinated activity to be ex-
858 plained.

859 **9 Discussion**

860 Two things have been established here, and they are of different kinds. The
861 constraints of Section 2 hold outright: synchrony does not deliver compatible
862 content, the resultant does not determine the state, a unique fixed point is
863 not a reconstruction, a region that cannot resolve a family cannot encode it,
864 and agreement about a change takes communication. Any account that reads
865 unity off phase coherence has to meet them, this one included. The account
866 set out in Section 1 is conditional at every step by contrast: each connection
867 in it is a hypothesis supplied rather than a result derived, and Table 1 names
868 all eight. Four of the ingredients assembled there are not standard — the

869 installed-energy condition, which makes a resource requirement necessary for
870 coherence on the stationary branch; the patch-local form of the coherence-
871 to-content bound, which is what keeps it from being vacuous; the stability
872 criterion and amplitude condition that settle when a low resultant reports a
873 winding field rather than weak coupling; and a machine-checked composition
874 carrying a witness that discharges all eight hypotheses at once, which is what
875 makes their conjunction non-empty rather than a list no object satisfies.
876 What follows takes the conditions in turn, as commitments the proposal
877 incurs, as measurements that would bear on them, and as questions the
878 framework leaves open.

879 **9.1 Unity and a minimal self as empirical commit-** 880 **ments**

881 We propose the state glued from compatible local descriptions as a corre-
882 late of conscious unity, and its accurate internal reconstruction as a correlate
883 of a minimal self. The motivations are that different contents belong to
884 one experienced situation and that the situation is experienced from a per-
885 spective. The conditional results identify relations a realization would need:
886 sufficient installed resources for the specified phase model, agreement about
887 shared content, and an encoding that supports reconstruction at the required
888 fidelity. Discharging E12–E89 for cortex would make the composition a state-
889 ment about cortex: that a cortical field above threshold relaxes to a cover

890 whose compatible local descriptions glue, and that the glued state is the
891 unique fixed point of a reconstruction map realized there. It would leave the
892 two identifications above exactly where they are, because neither is among
893 the eight and neither follows from their conjunction; they are the commit-
894 ment this subsection names, and Section 9.2 lists the three observations that
895 would count against it.

896 Compatibility allows different descriptions to contribute different infor-
897 mation. It also permits uncertainty and conflicting judgments: apparent cup
898 position and judged cup position can be distinct quantities in a consistent
899 description of their discrepancy. Which quantities are shared must be de-
900 termined independently. Choosing them solely to secure agreement would
901 deprive the proposal of empirical force.

902 The minimal-self proposal concerns the perceiver’s position and condition
903 within the situation, such as the cup’s location relative to a body that can
904 reach it. The encoding could be implicit and need carry no autobiograph-
905 ical narrative. Its proposed necessity is philosophical: even an unreflective
906 episode has a perspective. Gluing by itself requires no self-encoding, and re-
907 constructability alone supplies no functional role. Altering the encoded hand
908 or viewpoint while holding the external scene fixed should change estimates
909 of the perceiver’s relation to the cup as the readout predicts. Whether that
910 represented perspective is experienced remains a further question.

9.2 What a test of the full proposal must establish

Both uniqueness results concern specified objects. They choose neither the cover nor the encoding, and do not select between candidate subjects. Integrated information theory treats exclusion explicitly (Tononi et al., 2016); here selection requires independent evidence about neural organization. A constant readout can contract and fix a predetermined state while discarding its input. The family criterion in Section 5.2 tests the sensitivity that this construction lacks, provided the relevant states and fidelity are declared independently.

A test must separately measure coupling relative to diffusion, content compatibility and self-reconstruction, alongside an independent endpoint for experience. Meeting all three without a unified episode would challenge sufficiency. A unified episode with incompatible descriptions would challenge the unity identification; one lacking the specified self-encoding would challenge the reflexive requirement. Awakening, dreaming and dissociative states offer possible settings, but absence of report cannot by itself define absence of experience.

The next physical requirement is a calibrated geometry–coupling relation and resolved phase dynamics. The installation bound contributes a necessary resource condition where its hardware representation applies; cortical mode prices and field-mediated coupling require measurement. For the representational connection, an informative experiment asks whether a controlled change in coordination alters independently decoded content agreement, with

934 controls for sensory information and decoder quality. The constructed-data
935 estimator establishes feasibility for specified contents. Neural local descrip-
936 tions, shared overlap encoders and the self-encoding have yet to be identified.
937 The population-size dependence of the phase-to-content bound further limits
938 its quantitative use, in the form Section 4.2 reads off the sheet.

939 The separation results explain why these measurements must be inde-
940 pendent. A stationary distribution leaves several recovery mechanisms indis-
941 tinguishable. Equal phase order can accompany incompatible descriptions
942 or different task information. The tested plasticity rule preserves coherence
943 while losing its designated template. Each finding concerns the specified
944 model and observable; their role is to delimit the evidence needed for the
945 proposed connections.

946 **9.3 What the conditions constrain in a digital candi-** 947 **date**

948 The results of Section 2 hold for any candidate, and for a large language
949 model (LLM) running on graphics processing units (GPUs) three of them are
950 checkable rather than merely stateable. That is where this framework bears
951 on claims made for such systems. It does not exclude them, and no result
952 here could; what it does is fix what a claim of unity or self-representation
953 commits its maker to measuring.

954 **Codes, not reports.** Reconstruction across a family is a counting con-
955 straint before it is anything else. If two macrostates the candidate is said
956 to represent satisfy $E(s) = E(t)$ while $d(s, t) > 2\varepsilon$, Eq. (10) rules out re-
957 constructing both within ε , whatever the readout. A system’s reports are
958 readouts. Identical verbal reports across two conditions are therefore consis-
959 tent with identical internal codes and settle nothing about the states, while
960 the converse claim — that the system represents its own condition across a
961 declared family — asserts that its internal code separates that family, which
962 is a statement about activations under intervention and not about text. The
963 measurement follows from the statement: declare the family and the fidelity
964 first, then test whether the encoding region distinguishes its members.

965 **The region, not the readout.** Where the encoding is read as the state’s
966 restriction to a region of the machine, the demand falls on the region before
967 any readout exists. For a decoder-only transformer the causal mask makes
968 the network an instance of the update model of Section 5: the activation at
969 position t and layer ℓ reads positions at most t at layer $\ell - 1$ and nothing
970 else, so a region’s causal past is fixed by the architecture rather than chosen
971 by interpretation. Two interventions outside that past that demand reports
972 more than 2ε apart cannot both be answered within ε from that region.
973 A claim that a model reports on its own processing consequently carries an
974 obligation to say which positions and layers the report’s causal past contains,
975 and a report that appears to track an intervention outside it indicates a

976 confound or a different region, not a faster channel.

977 **Agreement about what.** Unity requires the candidate’s local descriptions
978 to agree on independently identified shared quantities (Section 4). Common
979 timing, a shared residual stream and fluent output supply none of that, for
980 the reason Section 2 gives: coordination bounds content disagreement only
981 through encoders already related on overlaps, and which quantities two parts
982 of a model describe in common is a decoding question carrying the failure
983 modes of Section 7. The cover is a choice, and choosing it to secure agreement
984 empties the claim.

985 **What is not excluded.** Digital representations can satisfy compatibility,
986 so finite hardware is compatible with the gluing construction and none of
987 the above is a no-go for machines. The coupling-gated route E56–E78 is a
988 different matter: it requires a calibrated physical coupling, its noise regime
989 and a connection from coordination to content agreement, and an attention
990 matrix or a shared clock supplies none of these identifications. Comput-
991 ing a field model likewise leaves its realization in the computing substrate
992 to be established. Those physical obligations fall on the cortical proposal
993 identically.

994 The installed-energy condition of Section 3.2 is restricted in the same
995 way. It constrains a macroscopic candidate only through κ , which is hard-
996 ware data supplied by mode profiles and prices, and Supplemental Material
997 reads the condition against a cortical metabolic budget to return the conver-

998 sion factor it would require rather than a verdict. The hardware comparisons
999 there have similarly restricted scope: fixed wiring can impose a correlation
1000 disadvantage, and point-supported kernels vanish under an atomless contin-
1001 uum measure, but neither comparison makes the representation conditions
1002 unattainable on GPUs. A universal exclusion would need a necessary condi-
1003 tion for consciousness and a proof that every relevant implementation violates
1004 it, and the present composition supplies neither.

1005 Whether experience requires a living substrate is the explicit subject of
1006 biological naturalism (Seth, 2026a,b). On that view the properties ground-
1007 ing experience are those of self-sustaining organisms, and simulating them
1008 does not secure them. The conditions stated here name no biological mech-
1009 anism, but computation described as such does not satisfy them either: a
1010 candidate must exhibit a physically realized coupling in a measured noise
1011 regime, agreement about independently identified shared quantities, and an
1012 internal reconstruction meeting a declared fidelity. Whether a realization
1013 of those conditions that is not alive would be accompanied by experience
1014 is the point at issue, and these results do not settle it. A related position
1015 locates the difference in the mode of computation rather than in life as such:
1016 biological systems compute across scales that do not separate, and along-
1017 side discrete operations carry continuous-valued dynamics belonging to their
1018 physical medium (Milinkovic & Aru, 2026). The coupling condition stated
1019 here is one measurable instance of that requirement, fixed to a physical field
1020 in a declared noise regime, and a description of such a field does not satisfy

1021 it.

1022 The neuroscientific case against *human-like* consciousness in text-only
1023 LLMs concerns embodied sensory interaction, thalamocortical organization
1024 and biological regulation linking perception and action to survival (Aru et al.,
1025 2023). Their absence supports skepticism, conditional on their causal rele-
1026 vance to experience. Those mechanisms are not requirements of the abstract
1027 gluing and reconstruction definitions, and workspace-like representations re-
1028 ported in some LLMs (Gurnee et al., 2026) supply candidate organization to
1029 investigate. What this framework adds to that investigation is an order for
1030 the questions: declare the family of states and the fidelity, fix the encoding
1031 region and read its causal past off the architecture, and only then ask what
1032 the system reconstructs. A claim that survives those three steps is a claim
1033 about the machine; one that does not survive them was a claim about its
1034 output.

1035 9.4 Explanatory scope

1036 Integrated-information, global-workspace and predictive-processing accounts
1037 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
1038 1997; Dehaene et al., 2014; Friston, 2010). The present proposal contributes a
1039 candidate physical interaction and conditional relations between local content
1040 and self-encoding. Gluing and reconstruction permit multiple substrates;
1041 the electromagnetic commitment enters through physical calibration. The
1042 formalization checks implications under their hypotheses, and its witnesses

1043 establish realizability on declared models. Two things stand outside that
1044 result and are not supplied by it: the eight hypotheses are discharged for
1045 toy components rather than for cortex, and the identification of the glued
1046 state with unity and of its reconstruction with a minimal self is a further
1047 commitment that the eight do not contain.

1048 The explanatory question is whether the specified relations occur in neural
1049 activity and matter for experience. In the cup example, this means establish-
1050 ing how coordination helps maintain one represented situation and how the
1051 perceiver’s place within it contributes to the episode. The proposal identi-
1052 fies candidate correlates and tests of that identification. Why those relations
1053 would be accompanied by experience remains unexplained.

1054 **Data and code availability**

1055 The Lean development, simulation scripts, tests and saved numerical sum-
1056 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`
1057 `cortical_coherence`. Publication simulation values are generated from
1058 saved summaries without rerunning production sweeps. The exploratory
1059 EEG analysis uses OpenNeuro dataset ds005620 (Bajwa et al., 2025); pre-
1060 processing and selection are described in Supplemental Material. The repos-
1061 itory contains the reproducible analysis and does not redistribute the raw
1062 recordings.

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1066 **Use of AI tools**

1067 AI assistance to this work was substantial and is not separable by task. Sev-
1068 eral coding assistants and chat interfaces were used in an interleaved way
1069 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
1070 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,
1071 Luna and Sol), Claude Code (Claude Opus 5), Gemini Pro through its web in-
1072 terface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter,
1073 accessed with the Hermes CLI. Between them they contributed to the Lean
1074 development, the simulation and analysis code, and the drafting and revision
1075 of both documents. Because the tools were used in combination and in alter-
1076 nation, no section, proof, figure or result is attributable to any one of them,
1077 and none is claimed to be. The author is responsible for all content, including
1078 everything a tool produced. Machine checking establishes the stated Lean
1079 results under their hypotheses; it does not validate biological interpretations
1080 or replace scientific review of generated code and prose.

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